Cross-state vector coherence

Condition

00 R1 / deepcoder
01 R1 / deepscaler
02 R1 / justrl
03 R1 / minimath
04 R1 / brorl
05 R1 / nemotron
06 R1 / light
07 Qwen / grpo-kl
08 Qwen / SFT thinking
09 Qwen / e3
10 Qwen / SFT non-thinking
11 Llama / coreward-i
12 Llama / gt
13 Llama / arithmetic-full
14 Qwen / controlled-RAW
15 Qwen / controlled-CCS-rho0
16 Qwen / controlled-CCS-rho1
19 MiMo / MiMo-RL-from-SFT
20 MiMo / Zero from SFT [branch]

0.0 0.2 0.4 0.6 0.8 1.0

Norm of summed gradients / sum of gradient norms

Alignment of missing off

Same condition order

-10^{-2} -10^{-3} -10^{-4} 0 10^{-4} 10^{-3} 10^{-2}

Cosine with descent (positive = locally helpful)

MATH CODE IF

Endpoint, fixed initial-student states; 2 heldout questions x first/last position per domain